STABLE MATCHING AND A TWO-TO-ONE MAP FOR TUNNELL'S TERNARY FORMS

AAKASH GURUNG AND KYUNGYONG LEE

Abstract. For every odd positive squarefree congruent number n, we construct a deterministic map $\Phi_n : \mathcal{B}_n \to \mathcal{A}_n$ whose fibers have cardinality two, where $\mathcal{B}_n$ and $\mathcal{A}_n$ are the representation sets of n by $2x^2+y^2+8z^2$ and $2x^2+y^2+32z^2$. Tunnell's theorem supplies the established identity $|\mathcal{B}_n| = 2|\mathcal{A}_n|$; more generally, the construction applies to every odd positive squarefree n satisfying this identity. Halving the third coordinate gives the even branch. On the odd branch, three integral quarter-turns define a partial bijection. We complete it by passing the residual points to their antipodal orbits and applying stable matching, with preferences determined by primitive midpoint directions. Equal-rank edges are disjoint, so the stable matching is unique. Its preference lists can be generated arithmetically from rank-two lattice cosets, and the proposal counts form a parking function with total at most $m(m+1)/2$, where m is the number of residual orbits on either side.

1. Introduction

Equalities between finite representation numbers naturally raise a finer combinatorial question: can the individual representations be matched in a structured way? We study this question for the ternary quadratic forms that occur in Tunnell's work on congruent numbers. For an odd positive integer n, let

$$\mathcal{B}_n = \{(x, y, z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = n\}, \tag{1}$$

$$\mathcal{A}_n = \{(u, v, w) \in \mathbb{Z}^3 : 2u^2 + v^2 + 32w^2 = n\}. \tag{2}$$

A ternary quadratic form is a homogeneous quadratic polynomial in three variables, and a representation of n is simply an integer triple at which the form has value n. We call n *squarefree* if no prime square divides it. Both sets above are finite because each squared coordinate is bounded by n.

A positive integer is *congruent* if it is the area of a right triangle with rational side lengths. Equivalently, the elliptic curve $y^2 = x^3 - n^2x$ has a rational point of infinite order; see [6, Chap. I, §9] and [9, pp. 323–324].

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A consequence of Tunnell's theorem is that every odd positive squarefree congruent number satisfies

$$|\mathcal{B}_n| = 2|\mathcal{A}_n|. \tag{3}$$

This consequence is the only input from Tunnell's theorem used below; no modular-form or elliptic-curve argument enters the construction itself.

Example 1.1 (The balance for $n = 41$). The equations themselves make this first example finite and elementary. For $n = 41$, a solution in $\mathcal{B}_{41}$ must satisfy $|x| \leq 4$, $|y| \leq 6$, and $|z| \leq 2$; a solution in $\mathcal{A}_{41}$ must satisfy the same first two bounds and $|w| \leq 1$. Sorting by the last coordinate gives

$\mathcal{B}_{41} : z$	0	1	−1	2	−2	total
number of (x, y)	4	8	8	6	6	32

and

$\mathcal{A}_{41} : w$	0	1	−1	total
number of (u, v)	4	6	6	16.

Thus the required 2:1 balance holds. For a single visible instance,

$$(2, -1, 2) \in \mathcal{B}_{41}, \qquad (2, -1, 1) \in \mathcal{A}_{41},$$

and the even-branch map simply halves the last coordinate.

Methods such as those of Bach and Ryan [2] compute the representation numbers in this identity. Our purpose is different. We use (3), established by Tunnell for the congruent numbers under consideration, and seek a structured map between the representations that it counts. Isolating the balance also gives a slightly more general construction for any odd positive squarefree n satisfying the same identity. We neither reprove Tunnell's identity nor propose a new congruent-number criterion.

Separate $\mathcal{B}_n$ according to the parity of its third coordinate:

$$\mathcal{B}_n^{\text{even}} = \{(x, y, z) \in \mathcal{B}_n : z \equiv 0 \pmod 2\},$$
$$\mathcal{B}_n^{\text{odd}} = \{(x, y, z) \in \mathcal{B}_n : z \equiv 1 \pmod 2\}.$$

The map

$$(x, y, 2w) \longmapsto (x, y, w) \tag{4}$$

is a bijection $\mathcal{B}_n^{\text{even}} \xrightarrow{\sim} \mathcal{A}_n$. Consequently (3) is equivalent to $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$, and the essential problem is to construct a bijection from the odd branch to the target set.

The residual problem is naturally bipartite. After a direct partial bijection is removed and the remaining points are passed to their $\{\pm 1\}$-orbits, every source orbit is joined to every target orbit. A primitive midpoint direction assigns a rank to each edge. Both endpoints order their incident edges by this rank, while equal-rank edges are vertex-disjoint. This is an instance of stable marriage with globally ranked pairs in the sense of Abraham, Levavi, Manlove, and O'Malley [1]. Its special edge structure forces a unique stable matching.

Stable matching and parking functions are classical combinatorial objects [4, 5, 7, 8]. Here they arise from the arithmetic geometry of two representation sets. The direct edges come from integral orthogonal transformations, whereas the residual edge ranking comes from primitive midpoint lines. Although balance alone would permit an arbitrary pairing, these arithmetic choices determine the map after the orientation conventions below have been fixed. The construction is equivariant under the antipodal involution $X \mapsto -X$, and the same edge data determine its inverse.

Theorem 1.2 (Two-to-one map under the Tunnell balance)**.** *Let n be an odd positive squarefree integer satisfying $|\mathcal{B}_n| = 2|\mathcal{A}_n|$. Fix the orientation and ordering conventions introduced below. Then the construction defines a deterministic map*

$$\Phi_n : \mathcal{B}_n \longrightarrow \mathcal{A}_n$$

such that every element of $\mathcal{A}_n$ has exactly two preimages. On $\mathcal{B}_n^{\mathrm{even}}$, the map is (4)*. On $\mathcal{B}_n^{\mathrm{odd}}$, it is the disjoint union of three quarter-turn branches and the unique stable matching of the remaining orbit sets. The later arithmetic generator and deferred-acceptance procedure compute this matching without first storing its complete edge table.*

Here a *fiber* is the set of all source points mapping to one target. The architecture of the theorem is the following. The top row gives one preimage of every target, and the bottom row gives the other:

$$\begin{aligned} &\mathcal{B}_n^{\mathrm{even}} \xrightarrow[\text{bijection}]{(x,y,2w)\mapsto(x,y,w)} \mathcal{A}_n, \\ \mathcal{B}_n^{\mathrm{odd}} = D_n \sqcup \mathcal{B}_{n,\circ}^{\mathrm{odd}} &\xrightarrow[\text{bijection}]{\Lambda_n \sqcup \Psi_n^{\circ}} I_n \sqcup \mathcal{A}_{n,\circ} = \mathcal{A}_n. \end{aligned} \tag{5}$$

The residual arrow in the bottom row is built through

$$\mathcal{B}_{n,\circ}^{\mathrm{odd}} \xrightarrow{\iota_B} \mathcal{S}_n^{\circ} \longrightarrow \overline{\mathcal{S}}_n \xrightarrow{\overline{M}_n} \overline{\mathcal{T}}_n \longrightarrow \mathcal{T}_n^{\circ} \xrightarrow{\iota_A^{-1}} \mathcal{A}_{n,\circ}. \tag{6}$$

The unlabeled arrows pass to antipodal orbits and then lift back with a specified sign. The cardinality hypothesis is used to prove that $\overline{M}_n$ is perfect; it is not used to choose the edge ranks.

Theorem 1.3 (Congruent-number case)**.** *For every odd positive squarefree congruent number n, the construction defines a deterministic map $\Phi_n : \mathcal{B}_n \to \mathcal{A}_n$ such that every element of $\mathcal{A}_n$ has exactly two preimages.*

Section 2 introduces the two coordinate models, and Section 3 constructs the direct branches. Section 4 constructs the unique residual matching, and Section 5 immediately assembles the two-to-one map. Sections 6–9 then show how to compute the residual matching and bound its proposal phase. We finish with the full $n = 41$ residual table and two further questions.

The examples at $n = 41$ follow two complementary lanes, not one point through both odd-branch mechanisms:

lane	odd source	target	even source
direct quarter-turn	$(-4, -1, 1)$	$(2, -1, 1)$	$(2, -1, 2)$
residual matching	$(-2, -5, 1)$	$(4, -3, 0)$	$(4, -3, 0)$.

Each lane therefore ends in a complete two-element fiber, but the two odd mechanisms act on disjoint subsets.

2. Coordinate models for the direct and residual constructions

We use one coordinate model for the direct local branches and another for midpoint directions. They are parallel descriptions of the original sets, not successive changes of coordinates:

$$\begin{array}{ccccccc} \mathcal{B}_n^{\mathrm{odd}} & \underset{e_B^{-1}}{\overset{e_B}{\longleftrightarrow}} & L_1(n) & \xrightarrow{J_1, J_2, J_3} & L_0(n) & \underset{e_A}{\overset{e_A^{-1}}{\longleftrightarrow}} & \mathcal{A}_n, \\ \mathcal{B}_n^{\mathrm{odd}} & \underset{\iota_B^{-1}}{\overset{\iota_B}{\longleftrightarrow}} & \mathcal{S}_n & \underset{\text{midpoint lines}}{\longleftrightarrow} & \mathcal{T}_n & \underset{\iota_A}{\overset{\iota_A^{-1}}{\longleftrightarrow}} & \mathcal{A}_n. \end{array}$$

The first row makes quarter-turns integral. The second puts sources and targets on the same quadratic surface so that their half-sums and half-differences are integral.

For reference, the main set and map symbols used below are collected here:

D_n, I_n, Λ_n	direct odd-source domain, its target image, and the quarter-turn bijection between them;
$\mathcal{B}_{n,\circ}^{\mathrm{odd}}, \mathcal{A}_{n,\circ}, \Psi_n^{\circ}$	points left after the direct layer and their residual bijection;
$\mathcal{S}_n, \mathcal{T}_n$	midpoint-coordinate lifts of the odd source and target sets;
$\overline{\mathcal{S}}_n, \overline{\mathcal{T}}_n$	antipodal residual orbit sets;
$\mathcal{P}_Q$	canonically oriented primitive midpoint directions;
Ψ_n, Φ_n	the full odd-branch bijection and the final two-to-one map.

2.1. Sum-of-three-squares coordinates for the direct map.

For $\epsilon \in \{0, 1\}$, put

$$L_\epsilon(n) = \{(a, b, c) \in \mathbb{Z}^3 : a^2 + b^2 + c^2 = n, \\ a \equiv b \pmod 4, \quad \frac{a-b}{4} \equiv \epsilon \pmod 2\}.$$

Lemma 2.1 (Sum-of-three-squares coordinates)**.** *The maps*

$$e_B(x, y, z) = (x + 2z, x - 2z, y), \qquad e_A(u, v, w) = (u + 4w, u - 4w, v)$$

are bijections

$$e_B : \mathcal{B}_n^{\mathrm{odd}} \xrightarrow{\sim} L_1(n), \qquad e_A : \mathcal{A}_n \xrightarrow{\sim} L_0(n),$$

with inverses

$$e_B^{-1}(a,b,c) = \left(\frac{a+b}{2}, c, \frac{a-b}{4}\right),$$
$$e_A^{-1}(a,b,c) = \left(\frac{a+b}{2}, c, \frac{a-b}{8}\right).$$

Proof. The norm identities are immediate:

$$(x+2z)^2 + (x-2z)^2 + y^2 = 2x^2 + y^2 + 8z^2,$$
$$(u+4w)^2 + (u-4w)^2 + v^2 = 2u^2 + v^2 + 32w^2.$$

The congruence conditions record whether $(a-b)/4$ is odd or even. Conversely, if $(a,b,c) \in L_1(n)$, then $(a+b)/2$ and $(a-b)/4$ are integral, the latter is odd, and the first norm identity gives a point of $\mathcal{B}_n^{\mathrm{odd}}$. If $(a,b,c) \in L_0(n)$, then $(a-b)/8$ is integral and the second identity gives a point of $\mathcal{A}_n$. The displayed inverse formulas therefore give inverses on both sides. □

Example 2.2 (The two coordinate changes at $n = 41$). Consider the odd source $X = (-4,-1,1) \in \mathcal{B}_{41}^{\mathrm{odd}}$. Lemma 2.1 sends it to

$$e_B(X) = (-2,-6,-1), \qquad (-2)^2 + (-6)^2 + (-1)^2 = 41.$$

Here $(-2) - (-6) = 4$, so $(a-b)/4 = 1$ is odd, as required for $L_1(41)$. Likewise, for $Y = (2,-1,1) \in \mathcal{A}_{41}$,

$$e_A(Y) = (6,-2,-1), \qquad 6^2 + (-2)^2 + (-1)^2 = 41,$$

and $(6-(-2))/4 = 2$ is even, as required for $L_0(41)$. The inverse formulas recover X and Y from these triples.

2.2. **Midpoint coordinates for the residual graph.** Put

$$Q(X_1,X_2,X_3) = 2X_1^2 + X_2^2 + 2X_3^2$$

and write

$$\langle X, Y\rangle_Q = 2X_1Y_1 + X_2Y_2 + 2X_3Y_3$$

for its associated bilinear form. Thus $Q(X+Y) = Q(X) + 2\langle X,Y\rangle_Q + Q(Y)$. Define

$$\mathcal{S}_n = \{E \in \mathbb{Z}^3 : Q(E) = n,\ E_3 \equiv 2 \pmod 4\},$$
$$\mathcal{T}_n = \{F \in \mathbb{Z}^3 : Q(F) = n,\ F_3 \equiv 0 \pmod 4\}.$$

The lifts

$$\iota_B(x,y,z) = (x,y,2z), \qquad \iota_A(u,v,w) = (u,v,4w)$$

identify $\mathcal{B}_n^{\mathrm{odd}}$ with $\mathcal{S}_n$ and $\mathcal{A}_n$ with $\mathcal{T}_n$.

Lemma 2.3 (Midpoint parity). *If $E \in \mathcal{S}_n$ and $F \in \mathcal{T}_n$, then*

$$\frac{E+F}{2}, \qquad \frac{E-F}{2}$$

are integral vectors with odd third coordinate.

Proof. Since n is odd, the second coordinates of both E and F are odd. Their third coordinates are congruent to 2 and 0 modulo 4, respectively. Consequently, modulo 8 one has

$$n \equiv 2E_1^2 + 1 \equiv 2F_1^2 + 1,$$

so E_1 and F_1 have the same parity. The second-coordinate halves are integral because both entries are odd, and the third-coordinate halves are odd because $E_3 \equiv 2 \pmod 4$ and $F_3 \equiv 0 \pmod 4$. Thus both displayed vectors are integral and have odd third coordinate. $\square$

For $0 \neq v \in \mathbb{Z}^3$, define

$$\operatorname{cont}(v) = \gcd(|v_1|, |v_2|, |v_3|).$$

Call v *primitive* if $\operatorname{cont}(v) = 1$. Let $\mathcal{P}_Q$ contain the unique representative, with first nonzero coordinate positive, of every pair $\{d, -d\}$ of primitive vectors with d_3 odd. Order $\mathcal{P}_Q$ lexicographically by

$$\operatorname{key}(d) = \big(Q(d), d_1, d_2, d_3\big). \tag{7}$$

This direction convention differs from the point-orbit convention used later: directions have first nonzero coordinate positive, whereas an antipodal point orbit is represented by its lexicographically smaller point. Keeping the two rules explicit prevents their signs from being confused.

Lemma 2.4 (Primitive generator of the midpoint line)**.** *For every $E \in \mathcal{S}_n$ and $F \in \mathcal{T}_n$, there are unique $d \in \mathcal{P}_Q$ and a nonzero odd integer q such that*

$$\frac{E+F}{2} = qd.$$

They satisfy

$$\langle E, d\rangle_Q = qQ(d), \qquad F = -E + 2qd.$$

For a fixed d, the edges $E \leftrightarrow F$ obtained in this way are pairwise disjoint.

Proof. By Lemma 2.3, the midpoint is a nonzero integral vector with odd third coordinate. Dividing by its content and applying the orientation convention gives q and d, with q odd. If $h = (E-F)/2$, equality of the norms gives $\langle d, h\rangle_Q = 0$, and hence $\langle E, d\rangle_Q = qQ(d)$ and $F = -E + 2qd$. Uniqueness follows from the unique sign-normalized primitive generator of $\mathbb{Q}(E+F) \cap \mathbb{Z}^3$, the integer points on the rational line spanned by $E+F$. For fixed E and d, the identity $q = \langle E, d\rangle_Q / Q(d)$ determines q, and then the formula determines F. Interchanging the roles of the endpoints gives the same conclusion at a target. Hence the edges belonging to a fixed direction are pairwise disjoint. $\square$

Example 2.5 (A midpoint direction at $n = 41$). Take the lifted residual points

$$E = (-2, -5, 2) \in \mathcal{S}_{41}, \qquad F = (4, -3, 0) \in \mathcal{T}_{41}.$$

Both have Q-value 41. Their half-sum and half-difference are

$$\frac{E+F}{2} = (1, -4, 1), \qquad \frac{E-F}{2} = (-3, -1, 1),$$

so Lemma 2.3 is visible directly: both vectors are integral and have odd third coordinate. The first vector is already primitive and canonically oriented, hence $q = 1$ and $d = (1, -4, 1)$. Moreover,

$$Q(d) = 20, \qquad \langle E, d\rangle_Q = 20 = qQ(d), \qquad -E + 2d = F.$$

The other midpoint line is generated, after sign normalization, by $e = (3, 1, -1)$, whose key begins with $Q(e) = 21$. Thus the key of d is the smaller of the two, explaining which representative-level edge is preferred when the construction later passes to antipodal orbits. Thus the midpoint line determines this edge exactly as in Lemma 2.4.

3. A direct quarter-turn correspondence

Define three integral rotations in the sum-of-three-squares model by

$$J_1(a, b, c) = (a, -c, b), \qquad J_2(a, b, c) = (c, b, -a), \qquad J_3(a, b, c) = (-b, a, c).$$

Each is an element of $\mathrm{SO}_3(\mathbb{Z})$: it is an integer matrix that preserves squared length and orientation. Geometrically, J_i fixes one coordinate axis and rotates the other two coordinates through a right angle.

Proposition 3.1 (Quarter-turn domains)**.** *Let $(a, b, c) \in L_1(n)$. Then*

$$\begin{aligned} J_1(a, b, c) \in L_0(n) &\iff a + c \equiv 0 \pmod 8, \\ J_2(a, b, c) \in L_0(n) &\iff c - b \equiv 0 \pmod 8, \\ J_3(a, b, c) \in L_0(n) &\iff a + b \equiv 0 \pmod 8. \end{aligned}$$

At most one of these three congruences holds.

Proof. The differences between the first two coordinates of the three images are $a + c$, $c - b$, and $-(a + b)$, respectively. Membership in $L_0(n)$ is therefore equivalent to the stated divisibility by 8.

In original odd-source coordinates, $c = y$ is odd and $a - b = 4z \equiv 4 \pmod 8$. If any two of the displayed congruences held, then, after eliminating a or b, one would obtain

$$a - b \equiv -2c \pmod 8.$$

Since c is odd, the right-hand side is 2 or 6 modulo 8, contradicting $a - b \equiv 4 \pmod 8$. Hence the three domains are pairwise disjoint. □

Transferring the rotations back to the original coordinates gives the following partial map. For $(x, y, z) \in \mathcal{B}_n^{\mathrm{odd}}$ with $x + 2z + y \equiv 0 \pmod 8$, set

$$\Lambda_1(x, y, z) = \left(\frac{x + 2z - y}{2},\ x - 2z,\ \frac{x + 2z + y}{8}\right). \tag{8}$$

If $y - x + 2z \equiv 0 \pmod 8$, set

$$\Lambda_2(x, y, z) = \left(\frac{x - 2z + y}{2},\ -x - 2z,\ \frac{y - x + 2z}{8}\right). \tag{9}$$

Finally, if $x \equiv 0 \pmod 4$, set

$$\Lambda_3(x,y,z) = \left(2z,\ y,\ -\frac{x}{4}\right). \tag{10}$$

The domains are pairwise disjoint by Proposition 3.1.

Theorem 3.2 (Direct partial bijection)**.** *The formulas* (8)–(10) *define a partial bijection*

$$\Lambda_n : D_n \xrightarrow{\sim} I_n$$

from a subset $D_n \subseteq \mathcal{B}_n^{\mathrm{odd}}$ to a subset $I_n \subseteq \mathcal{A}_n$. Its inverse is determined from a target $(u,v,w) \in \mathcal{A}_n$ by the following mutually exclusive conditions and formulas:

$$\begin{aligned}
u + 4w - v \equiv 4 \pmod 8 &: \quad \left(\frac{u+4w+v}{2}, -u+4w, \frac{u+4w-v}{4}\right),\\
u - 4w + v \equiv 4 \pmod 8 &: \quad \left(\frac{u-4w-v}{2}, u+4w, \frac{4w-u-v}{4}\right),\\
u \equiv 2 \pmod 4 &: \quad \left(-4w, v, \frac{u}{2}\right).
\end{aligned}$$

Moreover, Λ_n is equivariant under the antipodal involution $X \mapsto -X$.

Proof. The construction is $e_A^{-1} J_i e_B$ on each branch, so norm preservation and integrality follow from Proposition 3.1. The inverse formulas are $e_B^{-1} J_i^{-1} e_A$. Their conditions state exactly that the inverse image lies in $L_1(n)$. They are mutually exclusive: the first two cannot both hold because v is odd, and either of them is incompatible with $u \equiv 2 \pmod 4$ by parity. Direct composition gives the two identity maps. Negation equivariance follows because each J_i is linear and each domain predicate is sign invariant. $\square$

Example 3.3 (The three quarter-turn branches). The first two branches already occur at $n = 11$; the direct construction does not require the cardinality hypothesis. Direct substitution gives

branch	X	$e_B(X)$	$\Lambda_i(X)$
Λ_1	$(-1,-1,1)$	$(1,-3,-1)$	$(1,-3,0)$
Λ_2	$(-1,1,-1)$	$(-3,1,1)$	$(1,3,0)$.

For the first row, $a + c = 1 - 1 = 0$, so J_1 applies and $J_1(1,-3,-1) = (1,1,-3) = e_A(1,-3,0)$. For the second row, $c - b = 1 - 1 = 0$, so J_2 applies and $J_2(-3,1,1) = (1,1,3) = e_A(1,3,0)$. These calculations exhibit the first two domain tests and their images.

For the $n = 41$ direct lane, continue with $X = (-4,-1,1)$ from Example 2.2. Its sum-of-three-squares coordinates are $(a,b,c) = (-2,-6,-1)$. Since

$$a + b = -8 \equiv 0 \pmod 8,$$

Proposition 3.1 selects J_3 and neither of the other two quarter-turns. Indeed,

$$J_3(-2,-6,-1) = (6,-2,-1) = e_A(2,-1,1).$$

In the original coordinates this is exactly

$$\Lambda_3(-4,-1,1) = (2,-1,1).$$

The inverse condition is $u \equiv 2 \pmod 4$, and the third inverse formula gives $(-4w, v, u/2) = (-4,-1,1)$.

Set

$$\mathcal{B}_{n,\circ}^{\text{odd}} = \mathcal{B}_n^{\text{odd}} \setminus D_n, \qquad \mathcal{A}_{n,\circ} = \mathcal{A}_n \setminus I_n,$$

and let $\mathcal{S}_n^\circ = \iota_B(\mathcal{B}_{n,\circ}^{\text{odd}})$ and $\mathcal{T}_n^\circ = \iota_A(\mathcal{A}_{n,\circ})$. Under the hypothesis of Theorem 1.2,

$$|\mathcal{S}_n^\circ| = |\mathcal{T}_n^\circ|. \tag{11}$$

Indeed, $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$ and Λ_n removes equally many points from the two sides.

4. Globally ranked stable matching on antipodal orbits

The residual sets are invariant under the antipodal involution $X \mapsto -X$. Since $n > 0$, no point in either set equals its negative; this is what it means here for the action of $\{\pm 1\}$ to be free. Define the corresponding orbit sets

$$\overline{\mathcal{S}}_n = \mathcal{S}_n^\circ/\{\pm 1\}, \qquad \overline{\mathcal{T}}_n = \mathcal{T}_n^\circ/\{\pm 1\}.$$

By (11), these sets have the same cardinality; write

$$|\overline{\mathcal{S}}_n| = |\overline{\mathcal{T}}_n| = m.$$

Each orbit is an antipodal pair $[X] = \{X, -X\}$. Choose the lexicographically smaller of X and $-X$ as its canonical representative $\operatorname{pc}([X])$. Let

$$G_n = K_{\overline{\mathcal{S}}_n, \overline{\mathcal{T}}_n}$$

be the complete bipartite graph on the two orbit sets.

Fix $[E] \in \overline{\mathcal{S}}_n$ and $[F] \in \overline{\mathcal{T}}_n$, with canonical representatives E and F. Apply Lemma 2.4 to the two edges between chosen representatives $E \leftrightarrow F$ and $E \leftrightarrow -F$. Let their direction keys be $k_+(E,F)$ and $k_-(E,F)$.

Lemma 4.1 (Edge ranking on antipodal orbits)**.** *The two keys $k_+(E,F)$ and $k_-(E,F)$ are distinct. The quantity*

$$\kappa([E],[F]) = \min\{k_+(E,F), k_-(E,F)\}$$

is independent of the chosen representatives. There is a unique sign

$$\eta([E],[F]) \in \{\pm 1\}$$

such that the representative-level edge

$$E \longleftrightarrow \eta([E],[F])F$$

has key $\kappa([E],[F])$.

Proof. Write
$$\frac{E+F}{2} = qd, \qquad \frac{E-F}{2} = ce$$
with primitive oriented directions d, e. Equality of the norms gives $\langle d, e\rangle_Q = 0$. The vectors are nonzero, so positive definiteness implies $d \neq e$ and hence the keys are distinct. Replacing either representative by its negative interchanges the two signed midpoint lines, so their unordered pair of keys is unchanged. The smaller one determines the unique sign. ☐

Example 4.2 (Choosing and lifting one signed orbit edge)*.* Take the canonical representatives
$$E = (-2, -5, -2), \qquad F = (-4, -3, 0)$$
of the orbits later denoted by s_1 and t_1. There are two representative-level choices. For $E \leftrightarrow F$, the midpoint is
$$\frac{E+F}{2} = (-3, -4, -1) = -(3, 4, 1),$$
so its oriented primitive direction has key $(36, 3, 4, 1)$. For $E \leftrightarrow -F$, the midpoint is
$$\frac{E-F}{2} = (1, -1, -1),$$
whose key is $(5, 1, -1, -1)$. The second key is smaller. Therefore
$$\kappa([E], [F]) = (5, 1, -1, -1), \qquad \eta([E], [F]) = -1,$$
and the canonical source representative lifts to
$$E \longmapsto -F = (4, 3, 0).$$
For the other source point $-E$, multiplying both endpoints by -1 gives the other lift $-E \mapsto F$. Thus the orbit edge encodes both point-level edges and their inverse signs.

If the actual source is σE, where $\sigma \in \{\pm 1\}$, the lift of an orbit edge is
$$\sigma E \longmapsto \sigma\eta([E], [F])F. \tag{12}$$
The same rule, read from the target, recovers the source.

Proposition 4.3 (Equal-key classes are matchings)**.** *For a fixed key k, the edges of G_n with key k form a matching: no two share a source or target orbit.*

Proof. At a fixed signed endpoint, a primitive direction determines at most one opposite signed endpoint by Lemma 2.4. It remains to check that the two signs of an endpoint do not create two orbits. If a fixed direction sends F to $E = -F + 2qd$, then at the endpoint $-F$ the multiplier is $-q$ and the partner is
$$-(-F) + 2(-q)d = -E.$$

Thus changing the sign of one endpoint changes the sign of the other and does not change its orbit. Hence a fixed key occurs at most once at either endpoint of G_n. □

Example 4.4 (Two disjoint edges with the same key). The key

$$(5, 1, -1, -1)$$

occurs on the edge (s_1, t_1) computed in Example 4.2. For a second edge, take

$$E' = (-2, 5, -2) = \mathrm{pc}(s_3), \qquad F' = (0, -3, 4) = \mathrm{pc}(t_4).$$

Their midpoint is $(-1, 1, 1) = -(1, -1, -1)$, so this edge has the same key. The sources s_1, s_3 and targets t_1, t_4 are distinct. Hence the two equal-key edges are disjoint, exactly as Proposition 4.3 requires.

Every vertex ranks its incident edges by increasing κ. Proposition 4.3 implies that equal-key edges are vertex-disjoint, so every vertex has a strict preference order. A pair $(s, t) \notin M$ *blocks* a matching M if both s and t prefer each other to their partners in M, with being unmatched least preferred. A matching with no blocking pair is *stable*. Thus G_n is a stable-marriage instance with globally ranked pairs [4, 5, 1]: the two preference systems are induced by one global ranking of edges.

Proposition 4.5 (Unique stable matching)**.** *The graph G_n has a unique stable matching. It is obtained by processing the edge keys in increasing order and selecting every edge whose two endpoints are still unmatched. Since $|\overline{\mathcal{S}}_n| = |\overline{\mathcal{T}}_n|$, this stable matching is perfect.*

Proof. First run the stated greedy procedure. If an edge is not selected, then at least one of its endpoints was already matched when that edge was processed. The selected edge at that endpoint has smaller key; equality is impossible by Proposition 4.3. Thus an unselected edge cannot block, and the greedy matching is stable. It is also perfect: otherwise balance would leave an unmatched source and an unmatched target, but the edge joining them would have been selected when it was processed.

It remains to prove uniqueness. Consider the least key among the edges whose endpoints have not already been removed. Every edge of that key is forced in any stable matching: neither endpoint has a smaller edge in the remaining graph, so omission would make the two endpoints a blocking pair. Proposition 4.3 says that these edges have disjoint endpoints, and hence they can be selected and removed simultaneously. A removed endpoint prefers its selected edge to every edge processed later, and the restriction of a stable matching to the remaining vertices is stable. Induction therefore shows that every stable matching agrees with the greedy matching. □

Example 4.6 (A two-by-two globally ranked instance). Before returning to the arithmetic graph, consider two sources s_1, s_2 and two targets t_1, t_2. Give the four edges the global ranks

	t_1	t_2
s_1	1	3
s_2	2	4

with smaller ranks preferred. The greedy rule first selects (s_1, t_1). The edges of ranks 2 and 3 then meet an already matched vertex, so the next selected edge is (s_2, t_2). Hence

$$M = \{(s_1, t_1), (s_2, t_2)\}.$$

The only other perfect matching is not stable, because (s_1, t_1) blocks it. This is the simplest instance of Proposition 4.5.

5. Assembly of the two-to-one map

Let

$$\overline{M}_n : \overline{\mathcal{S}}_n \xrightarrow{\sim} \overline{\mathcal{T}}_n$$

be the unique stable matching of Proposition 4.5. Lift it to residual points using (12), and transfer back to the original coordinates. Denote the resulting residual bijection by

$$\Psi_n^\circ : \mathcal{B}_{n,\circ}^{\mathrm{odd}} \xrightarrow{\sim} \mathcal{A}_{n,\circ}.$$

Define the odd-branch bijection

$$\Psi_n(X) = \begin{cases} \Lambda_n(X), & X \in D_n, \\ \Psi_n^\circ(X), & X \in \mathcal{B}_{n,\circ}^{\mathrm{odd}}. \end{cases}$$

Proof of Theorem 1.2. The direct layer is a bijection $D_n \to I_n$ by Theorem 3.2. The residual orbit matching is perfect by Proposition 4.5, and the sign lift pairs the two points in each source orbit with the two points in the matched target orbit. Hence Ψ_n is a bijection $\mathcal{B}_n^{\mathrm{odd}} \to \mathcal{A}_n$.

Define

$$\Phi_n(x, y, z) = \begin{cases} (x, y, z/2), & z \text{ even}, \\ \Psi_n(x, y, z), & z \text{ odd}. \end{cases}$$

For $(u, v, w) \in \mathcal{A}_n$, the even branch contributes the preimage $(u, v, 2w)$, and the odd branch contributes the unique preimage $\Psi_n^{-1}(u, v, w)$. They are distinct because their third coordinates have opposite parity, and there are no further preimages on either branch. Thus every fiber has cardinality two. ☐

Proof of Theorem 1.3. For an odd squarefree congruent number n, Tunnell's theorem gives (3), so Theorem 1.2 applies. ☐

Remark 5.1 (Role of cardinality). The formulas defining the direct layer and edge keys make sense without (3). The equality is used only to make the two residual orbit sets equally large; Proposition 4.5 then makes their stable matching perfect. The later generator and deferred-acceptance procedure compute this already defined matching.

Example 5.2 (The direct $n = 41$ fiber). Let $Y = (2, -1, 1) \in \mathcal{A}_{41}$. The even branch gives

$$(2, -1, 2) \longmapsto (2, -1, 1),$$

while Example 3.3 gives the odd preimage

$$(-4, -1, 1) \longmapsto (2, -1, 1).$$

Their last coordinates have opposite parity, so

$$\Phi_{41}^{-1}(2, -1, 1) = \{(2, -1, 2), (-4, -1, 1)\}.$$

This is the complete fiber in the direct lane announced in the introduction.

6. Arithmetic generation of preference lists

The main theorem is already proved. We now explain how to compute the residual matching without first constructing all m^2 edges. For a fixed source orbit, its preference list can be recovered by searching for the primitive midpoint directions of its incident edges. We first bound this search and then write it as a problem in a rank-two affine lattice.

Lemma 6.1 (Bound for the preferred midpoint generator)**.** *If d is the direction realizing the smaller signed orientation of a residual orbit edge, then*

$$Q(d) \leq \frac{n-1}{2}.$$

Proof. The two midpoint decompositions from Lemma 4.1 give

$$E = qd + ce, \qquad F = qd - ce.$$

Since d and e are orthogonal, we also have

$$n = q^2 Q(d) + c^2 Q(e).$$

The chosen orientation gives $\mathrm{key}(d) < \mathrm{key}(e)$, hence $Q(d) \leq Q(e)$. Since q and c are nonzero,

$$n \geq Q(d) + Q(e) \geq 2Q(d).$$

The claim follows because n is odd. □

Example 6.2 (The midpoint bound is attained). In Example 2.5, the preferred direction satisfies

$$Q(d) = 20 = \frac{41-1}{2}.$$

Thus the bound in Lemma 6.1 can be an equality.

6.1. **An orthogonal-complement lattice.** We use the isometric embedding

$$\tau(X_1, X_2, X_3) = (X_1 + X_3, X_2, X_1 - X_3).$$

Then

$$\|\tau(X)\|^2 = Q(X), \qquad \tau(X) \cdot \tau(Y) = \langle X, Y \rangle_Q .$$

The image of τ is the index-two lattice

(13)

$$\mathcal{L}_\tau = \{r \in \mathbb{Z}^3 : r_1 \equiv r_3 \pmod 2\}, \qquad \tau^{-1}(r) = \left(\frac{r_1 + r_3}{2}, r_2, \frac{r_1 - r_3}{2}\right).$$

Let us fix a residual source orbit. We write E for its canonical lifted representative and set $p = \tau(E)$. Since $\|p\|^2 = n$ and n is squarefree, p is primitive; otherwise the square of a common divisor of its coordinates would divide n.

Before choosing coordinates, the geometric search can be written as follows. If $h = Q(d)$ and $t = qh$, then $r = \tau(d)$ is an integer point in

$$\underbrace{\{r : p \cdot r = t\}}_{\text{an affine plane}} \cap \underbrace{\{r : \|r\|^2 = h\}}_{\text{a sphere}} \cap \mathcal{L}_\tau. \tag{14}$$

The integer points parallel to this plane form the rank-two lattice $p^\perp \cap \mathbb{Z}^3$. We next choose coordinates on this lattice and use them to turn the plane-sphere intersection into a quadratic equation in two integers.

Lemma 6.3 (Oriented kernel basis and Bézout vector)**.** *There are deterministically computable vectors $e_1, e_2, z_0 \in \mathbb{Z}^3$ such that*

$$e_1 \times e_2 = p, \qquad p \cdot z_0 = 1.$$

The pair e_1, e_2 is an oriented basis of $p^\perp \cap \mathbb{Z}^3$ and may be Lagrange–Gauss reduced while preserving $e_1 \times e_2 = p$.

Proof. Since p is primitive, there is a Bézout vector z_0 with $p \cdot z_0 = 1$. Applying Smith normal form to the row vector p gives an integral basis e_1, e_2 of its kernel. Every $x \in \mathbb{Z}^3$ has the unique decomposition

$$x = \bigl(x - (p \cdot x)z_0\bigr) + (p \cdot x)z_0.$$

so e_1, e_2, z_0 form a basis of $\mathbb{Z}^3$. The vector $e_1 \times e_2$ is an integral multiple of the primitive normal vector p. Taking its scalar product with z_0 shows that the multiple is ± 1, and reversing the kernel basis if necessary gives $e_1 \times e_2 = p$. The usual integral Lagrange–Gauss steps

$$(e_1, e_2) \mapsto (e_1, e_2 - me_1), \qquad (e_1, e_2) \mapsto (e_2, -e_1)$$

have determinant 1, preserve the cross product, and terminate with $|2e_1 \cdot e_2| \leq \|e_1\|^2 \leq \|e_2\|^2$. $\square$

We use Smith normal form in its usual algorithmic form: integer column operations, obtained from the Euclidean algorithm, change the primitive row p to $(1, 0, 0)$. Transforming the last two standard basis vectors back gives a basis of $p^\perp \cap \mathbb{Z}^3$. We take nonnegative Euclidean remainders and use lexicographic tie breaking. For Lagrange–Gauss reduction, we round ties in the nearest-integer step toward zero and choose the lexicographically smaller oriented basis when the two reduced vectors have the same norm. These choices make the auxiliary frame deterministic. They do not affect the final preference list, which is sorted by the intrinsic direction key. We refer to [3, Chap. 2] for these standard lattice algorithms.

We write

$$\alpha = \|e_1\|^2, \quad \beta = e_1 \cdot e_2, \quad \gamma = \|e_2\|^2,$$

$$\delta = z_0 \cdot e_1, \quad \varepsilon = z_0 \cdot e_2, \quad \zeta = \|z_0\|^2.$$

The identity $e_1 \times e_2 = p$ gives

$$\alpha\gamma - \beta^2 = n. \tag{15}$$

This follows from

$$\alpha\gamma - \beta^2 = \|e_1 \times e_2\|^2 = \|p\|^2 = n.$$

6.2. **Affine norm equations.** For an edge incident to E, write its midpoint data as

$$\frac{E+F}{2} = qd, \qquad h = Q(d), \qquad r = \tau(d).$$

We then have

$$\|r\|^2 = h, \qquad p \cdot r = qh. \tag{16}$$

We set $t = qh$. The frame from the previous subsection now gives coordinates on the affine plane $p \cdot r = t$.

Lemma 6.4 (Parametrization of a lattice coset)**.** *Every integral solution of $p \cdot r = t$ is uniquely of the form*

$$r = tz_0 + \mu e_1 + \nu e_2, \qquad \mu, \nu \in \mathbb{Z}.$$

For fixed h, q, ν, the condition $\|r\|^2 = h$ is equivalent to

$$\alpha\mu^2 + 2(\beta\nu + t\delta)\mu + \gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h = 0. \tag{17}$$

Proof. The difference $r - tz_0$ is orthogonal to p, and e_1, e_2 form an integral basis of the orthogonal-complement lattice. Expanding the squared norm gives (17). □

To determine the possible values of ν, we write

$$K = \beta\delta - \alpha\varepsilon, \qquad M = h(n - q^2 h).$$

The discriminant of (17), after dividing by 4, is

$$\Delta_\nu = (\beta\nu + t\delta)^2 - \alpha(\gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h).$$

Lemma 6.5 (Short interval identity)**.** *One has*

$$n\Delta_\nu = \alpha M - (n\nu - tK)^2. \tag{18}$$

Put $R = \left\lfloor \sqrt{\alpha M} \right\rfloor$*. By* (18)*,* $\Delta_\nu \geq 0$ *exactly when* ν *lies in the interval*

$$\left\lceil \frac{tK - R}{n} \right\rceil \leq \nu \leq \left\lfloor \frac{tK + R}{n} \right\rfloor. \tag{19}$$

For each such ν*, let* $s = \left\lfloor \sqrt{\Delta_\nu} \right\rfloor$*. If* $s^2 = \Delta_\nu$*, the possible values of* μ *form the set*

$$\left\{ \frac{-(\beta\nu + t\delta) + u}{\alpha} : u \in \{s, -s\}, \quad \alpha \mid -(\beta\nu + t\delta) + u \right\}. \tag{20}$$

If $s^2 \neq \Delta_\nu$*, there are no roots. In particular, when* $\Delta_\nu = 0$*, the set* (20) *contains at most one integer rather than two copies of the same root.*

Proof. The Gram matrix of the basis e_1, e_2, z_0 is

$$G = \begin{pmatrix} \alpha & \beta & \delta \\ \beta & \gamma & \varepsilon \\ \delta & \varepsilon & \zeta \end{pmatrix}.$$

Since $\det[e_1\ e_2\ z_0] = 1$, one has $\det G = 1$. Expanding gives

$$n\zeta - (\alpha\varepsilon^2 - 2\beta\delta\varepsilon + \gamma\delta^2) = 1,$$

and a direct rearrangement using $n = \alpha\gamma - \beta^2$ gives

$$K^2 + n(\delta^2 - \alpha\zeta) = -\alpha.$$

The discriminant can now be written as

$$\Delta_\nu = -n\nu^2 + 2tK\nu + t^2(\delta^2 - \alpha\zeta) + \alpha h.$$

After multiplying by n and using the preceding identity together with $t = qh$, we obtain

$$n\Delta_\nu = \alpha(nh - t^2) - (n\nu - tK)^2 = \alpha h(n - q^2 h) - (n\nu - tK)^2,$$

which is (18). Since $n\nu - tK$ is integral, its square is at most αM precisely when $|n\nu - tK| \leq \left\lfloor \sqrt{\alpha M} \right\rfloor$; this is equivalent to (19). The quadratic formula gives (20), interpreted as a set. $\square$

Example 6.6 (Recovering one $n = 41$ edge arithmetically). Return to $E = (-2, -5, 2)$ and $F = (4, -3, 0)$ from Example 2.5. The isometry gives

$$p = \tau(E) = (0, -5, -4).$$

One convenient oriented kernel basis and Bézout vector are

$$e_1 = (1, 0, 0), \qquad e_2 = (0, -4, 5), \qquad z_0 = (0, -1, 1).$$

These vectors follow directly from the two defining equations. The kernel equation $p \cdot r = 0$ is $-5r_2 - 4r_3 = 0$, so $r_2 = -4k$ and $r_3 = 5k$, while r_1 is free. This gives e_1, e_2, and the Bézout identity $5 - 4 = 1$ gives z_0. Hence $e_1 \times e_2 = p$ and $p \cdot z_0 = 1$. For this frame,

$$\alpha = 1, \quad \beta = 0, \quad \gamma = 41,$$
$$\delta = 0, \quad \varepsilon = 9, \quad \zeta = 2, \qquad K = -9.$$

The midpoint direction is $d = (1, -4, 1)$, so

$$h = Q(d) = 20, \qquad q = 1, \qquad t = 20, \qquad M = 20(41 - 20) = 420.$$

Since $R = \left\lfloor \sqrt{420} \right\rfloor = 20$ and $tK = -180$, the interval (19) becomes

$$\left\lceil \frac{-200}{41} \right\rceil \leq \nu \leq \left\lfloor \frac{-160}{41} \right\rfloor,$$

which leaves only $\nu = -4$. The short interval identity then gives

$$41\Delta_{-4} = 420 - \big(41(-4) - 20(-9)\big)^2 = 164, \qquad \Delta_{-4} = 4.$$

Thus (20) gives $\mu = \pm 2$. The choice $\mu = 2$ produces

$$r = 20z_0 + 2e_1 - 4e_2 = (2, -4, 0), \qquad \tau^{-1}(r) = (1, -4, 1) = d.$$

The other root has negative first coordinate and is removed by the orientation convention. The outer loop also uses negative q, so the canonically oriented opposite direction is recovered there whenever it gives the preferred signed edge. Finally,

$$-E + 2d = (4, -3, 0) = F.$$

Since the canonical representative of $[F]$ is $F_0 = (-4, 3, 0)$, the generator records

$$\big((20, 1, -4, 1), [F_0], -1\big).$$

This recovers one entry of the preference list directly from the arithmetic of the source E.

6.3. **The ordered-adjacency generator.** The preceding calculation gives a finite procedure for producing the complete preference list of a fixed source orbit.

Algorithm 6.7 (Ordered-adjacency generator)**.** Fix a residual source orbit $[E]$ and let $E = \mathrm{pc}([E])$. We perform the following steps:

(1) let $h = 1, 2, \ldots, (n-1)/2$;
(2) let q range over the nonzero odd integers satisfying $q^2 h < n$, and put $t = qh$ and $M = h(n - q^2 h)$;
(3) let ν range over the interval (19);
(4) compute Δ_ν and the set of integral roots (20); for each root form $r = tz_0 + \mu e_1 + \nu e_2$;
(5) retain r only if $r_1 \equiv r_3 \pmod 2$ and $d = \tau^{-1}(r)$ is primitive, canonically oriented, and has odd third coordinate;
(6) form $F = -E + 2qd$ and retain it only if $F \in \mathcal{T}_n^\circ$; write $F = \eta F_0$, where $F_0 = \mathrm{pc}([F])$ and $\eta \in \{\pm 1\}$;

(7) compare d with the primitive oriented direction of $(E-F)/2$. Retain the record

$$(\mathrm{key}(d), [F_0], \eta)$$

precisely when d has the smaller key.

For each h, we place the surviving records in a finite set and sort them by (d_1, d_2, d_3). Since h increases in the outer loop, these blocks together form the strict preference list of $[E]$. The square roots and divisibility tests use only integer arithmetic. When the discriminant is zero, the two signs in the quadratic formula contribute only one root. The generator keeps one sorted h-block at a time, returns its records, and then moves to the next nonempty block.

Theorem 6.8 (Correctness of the ordered-adjacency generator)**.** *For every residual source orbit $[E]$, Algorithm 6.7 terminates and outputs every incident edge exactly once, in strictly increasing order of its key.*

Proof. We first check that every output is an incident edge. A retained record satisfies $Q(d) = h$ and $\langle E, d\rangle_Q = qh$, and hence

$$Q(-E + 2qd) = Q(E) - 4q\,\langle E, d\rangle_Q + 4q^2 Q(d) = n.$$

The test $F \in \mathcal{T}_n^\circ$ gives the required congruence and exclusion conditions, so the record defines a valid edge between the chosen representatives. The last comparison keeps the preferred sign of the target orbit, and η is the sign used in (12).

We next take an incident edge and choose its preferred signed lift from E. Lemma 6.1 gives the stated range for $h = Q(d)$. Its vectors p and $r = \tau(d)$ satisfy (16). By Lemma 6.4, r has unique integral coordinates μ, ν, and Lemma 6.5 shows that the procedure recovers them. The τ-lattice, primitivity, orientation, parity, and residual tests all hold for this lift, and the final comparison retains it.

It remains to check that no edge appears twice. The values h and d are intrinsic to a retained edge, while $q = \langle E, d\rangle_Q / Q(d)$ is then forced. The affine coordinates μ, ν are unique, and (20) is a set even when $\Delta_\nu = 0$. Hence no record is emitted twice. □

Example 6.9 (A complete generated preference row at $n = 41$). For $E = (-2, -5, 2)$, direct enumeration of the residual target set gives four target orbits. Write their canonical representatives as

$$\begin{aligned} F_1 &= (-4, -3, 0), & F_2 &= (0, -3, 4), \\ F_3 &= (-4, 3, 0), & F_4 &= (0, -3, -4). \end{aligned}$$

Running Algorithm 6.7 gives the following surviving signed midpoint for each orbit. The last column records the Q-value of the discarded midpoint direction obtained from the opposite sign.

orbit	η	$(E+\eta F_i)/2$	$\kappa([E],[F_i])$	other Q
$[F_1]$	-1	$(1,-1,1)$	$(5,1,-1,1)$	36
$[F_2]$	-1	$(-1,-1,-1)$	$(5,1,1,1)$	36
$[F_3]$	-1	$(1,-4,1)$	$(20,1,-4,1)$	21
$[F_4]$	1	$(-1,-4,-1)$	$(20,1,4,1)$	21

The corresponding affine data (h,q,μ,ν), in the same row order, are

$$(5,1,2,-1),\qquad (5,-1,2,1),\qquad (20,1,2,-4),\qquad (20,-1,2,4).$$

For example, the third row is the full calculation in Example 6.6; its opposite sign gives the direction $(3,1,-1)$ of Q-value 21. In the second and fourth rows the displayed midpoint is the negative of its canonically oriented primitive direction, which is why the key has positive first coordinate. Sorting the surviving records gives the entire strict preference row

$$[F_1],\ [F_2],\ [F_3],\ [F_4]$$

with keys in the order displayed. This is the complete preference row; the table also records the four opposite-sign candidates removed by the last comparison.

Remark 6.10 (Reduced-basis search-interval bound)*.* For a Lagrange–Gauss reduced basis,

$$|2\beta| \le \alpha \le \gamma.$$

Equation (15) gives $\alpha \le 2\sqrt{n/3}$. The interval (19) is therefore shorter than the corresponding unrestricted scan of the orthogonal-complement lattice. We will not use this observation in the correctness or matching-phase proofs.

7. Deferred acceptance with generated preferences

Proposition 4.5 describes the matching by a global greedy procedure. To avoid sorting all edge keys at once, we use the source-proposing Gale–Shapley algorithm [4] together with the preference generators above. Here an iterator means a stateful generator for one fixed preference list. It changes when an entry is computed, but not the order of the list.

Enumerate the finite residual orbit sets $\overline{\mathcal{S}}_n$ and $\overline{\mathcal{T}}_n$, and attach to each source $s \in \overline{\mathcal{S}}_n$ the ordered generator of Theorem 6.8. A target compares two proposals by their global edge rank.

Algorithm 7.1 (Source-proposing deferred acceptance with generated preferences)**.** Initially every source is free and every target tentatively holds no proposal. Place the sources in a first-in, first-out queue in lexicographic order of their canonical representatives. Repeatedly remove a free source s from the front of the queue and request the next record (k,t,η) from its ordered generator.

(1) If t tentatively holds no proposal, it tentatively holds (s,k,η).

(2) If t tentatively holds (s', k', η') and $k < k'$, then t replaces that record by (s, k, η) and s' is appended to the queue.
(3) Otherwise t rejects s, which is appended to the queue.

The algorithm ends when the queue is empty.

Example 7.2 (Deferred acceptance on two sources). Run Algorithm 7.1 on the ranks in Example 4.6, with initial queue (s_1, s_2). The proposals are

$$s_1 \longrightarrow t_1, \qquad s_2 \longrightarrow t_1 \quad \text{(rejected)}, \qquad s_2 \longrightarrow t_2.$$

The first proposal is held by t_1. The second has rank 2, so t_1 keeps its rank-1 proposal from s_1 and rejects s_2. The requeued source then proposes to t_2. The queue is empty and the output is $\{(s_1, t_1), (s_2, t_2)\}$, the unique stable matching found by the greedy argument.

Theorem 7.3 (Correctness of deferred acceptance)**.** *Algorithm 7.1 terminates and returns the unique perfect stable matching of Proposition 4.5. The stored sign η lifts every selected edge to a representative-level source-target pair. The inverse at a target is recovered by rerunning the same deterministic global computation; no proposal transcript is required.*

Proof. A source proposes in strict preference order and never proposes to the same target twice. We first check that a free source never exhausts its generator. If a free source s had proposed to all m targets, then every target would hold a proposal: once a target receives its first proposal, it remains occupied thereafter. The held proposals would come from m distinct sources, since a source can be held by at most one target and proposes again only after becoming free. None could come from the free source s, leaving only $m - 1$ possible sources, a contradiction. Thus every request made by the algorithm has a next record. Since there are only m^2 possible proposals, the algorithm terminates.

When the queue is empty, every source is held by a target. There are m sources and m targets, and no target holds two sources, so the resulting matching is perfect.

Suppose (s, t) blocked the final matching. Since s prefers t to its final partner, s proposed to t earlier. The target either rejected s immediately or later displaced it for a proposal of smaller key. A target's held key only decreases, so its final edge is smaller than (s, t), a contradiction. The output is stable and hence equals the unique stable matching of Proposition 4.5.

For the inverse, let F_0 be the canonical representative of a matched target orbit, and suppose its matched record is (E_0, F_0, η) with E_0 canonical. The lifted matching sends E_0 to ηF_0. Hence an actual target ρF_0, with $\rho \in \{\pm 1\}$, has preimage $\rho\eta E_0$. Lemma 4.1 makes η intrinsic to the two orbits, and rerunning the deterministic computation reconstructs the same matching and the same sign. This recovers the inverse at the given target without retaining the first run's proposal transcript. $\square$

Example 7.4 (Recovering a residual preimage). Take the actual lifted target $F = (4, -3, 0)$. Its canonical orbit representative is $F_0 = (-4, 3, 0)$, so $F = \rho F_0$ with $\rho = -1$. Rerunning the deterministic matching returns the record from Example 6.9 with

$$E_0 = (-2, -5, 2), \qquad \eta = -1.$$

The inverse sign formula gives

$$\rho\eta E_0 = (-1)(-1)E_0 = E_0.$$

Finally, $\iota_B^{-1}(E_0) = (-2, -5, 1)$. Thus the inverse at the original target is

$$(4, -3, 0) \longmapsto (-2, -5, 1).$$

The example makes explicit that "recovering the inverse" means rerunning the global deterministic matching and then applying a local sign calculation; it is not a separate pointwise formula.

8. The two $n = 41$ example lanes

The direct lane was completed in Examples 3.3 and 5.2:

$$(-4, -1, 1) \longmapsto (2, -1, 1) \quad \text{with even companion } (2, -1, 2).$$

We now complete the distinct residual lane

$$(-2, -5, 1) \longmapsto (4, -3, 0) \quad \text{with even companion } (4, -3, 0).$$

The two odd-source mechanisms are alternatives on disjoint subsets; they are not successive steps in one fiber.

Example 8.1 (The residual matching for $n = 41$). The residual graph for $n = 41$ has four source orbits $s_1, \ldots, s_4$ and four target orbits $t_1, \ldots, t_4$. We index them by the following canonical lifted representatives:

$$\begin{array}{ll}
\mathrm{pc}(s_1) = (-2, -5, -2), & \mathrm{pc}(s_2) = (-2, -5, 2), \\
\mathrm{pc}(s_3) = (-2, 5, -2), & \mathrm{pc}(s_4) = (-2, 5, 2), \\
\mathrm{pc}(t_1) = (-4, -3, 0), & \mathrm{pc}(t_2) = (-4, 3, 0), \\
\mathrm{pc}(t_3) = (0, -3, -4), & \mathrm{pc}(t_4) = (0, -3, 4).
\end{array}$$

In the next display, each row is in preference order and the bracketed four-tuple is the corresponding edge key:

$$\begin{array}{ll}
s_1: & t_1[5, 1, -1, -1],\ t_3[5, 1, 1, -1],\ t_2[20, 1, -4, -1],\ t_4[20, 1, 4, -1], \\
s_2: & t_1[5, 1, -1, 1],\ t_4[5, 1, 1, 1],\ t_2[20, 1, -4, 1],\ t_3[20, 1, 4, 1], \\
s_3: & t_4[5, 1, -1, -1],\ t_2[5, 1, 1, -1],\ t_3[20, 1, -4, -1],\ t_1[20, 1, 4, -1], \\
s_4: & t_3[5, 1, -1, 1],\ t_2[5, 1, 1, 1],\ t_4[20, 1, -4, 1],\ t_1[20, 1, 4, 1].
\end{array}$$

The complete generated row in Example 6.9 is exactly the s_2 row here: its F_1, F_2, F_3, F_4 are respectively t_1, t_4, t_2, t_3. Thus the abstract generator calculation and this graph table describe the same four edges. With the queue convention of Algorithm 7.1, the proposal sequence is

$$s_1 \longrightarrow t_1, \quad s_2 \longrightarrow t_1 \quad \text{(rejected)}, \quad s_3 \longrightarrow t_4,$$
$$s_4 \longrightarrow t_3, \quad s_2 \longrightarrow t_4 \quad \text{(rejected)}, \quad s_2 \longrightarrow t_2.$$

Thus the algorithm makes six proposals. Its lifted residual pairs are

$$\begin{aligned}(-2,-5,-2) &\longleftrightarrow (4,3,0),\\ (-2,-5,2) &\longleftrightarrow (4,-3,0),\\ (-2,5,-2) &\longleftrightarrow (0,-3,4),\\ (-2,5,2) &\longleftrightarrow (0,-3,-4).\end{aligned}$$

For instance,

$$(-2,-5,1) \in \mathcal{B}_{41}^{\mathrm{odd}} \longmapsto (4,-3,0) \in \mathcal{A}_{41}.$$

Thus

$$\Phi_{41}^{-1}(4,-3,0) = \{(-2,-5,1),(4,-3,0)\}.$$

9. Parking functions and matching-phase bounds

A sequence $(a_1,\dots,a_m)$ of positive integers is a *parking function* if its nondecreasing rearrangement $b_1 \le \cdots \le b_m$ satisfies $b_i \le i$ for $1 \le i \le m$; see [7, 8]. For $s \in \overline{\mathcal{S}}_n$, let $p(s)$ be the number of proposals made by s. Equivalently, $p(s)$ is the rank of its final target in its preference list. In the proof below, $M = \overline{M}_n$ denotes the final stable matching.

Theorem 9.1 (The proposal counts form a parking function)**.** *Order the sources $s_1,\dots,s_m$ so that the keys of their final matched edges are nondecreasing. Then*

$$p(s_i) \le i \qquad (1 \le i \le m).$$

Let $b_1 \le \cdots \le b_m$ be the nondecreasing rearrangement of the proposal counts. Then $b_i \le i$, so the proposal-count sequence is a parking function. Moreover,

$$\sum_{s\in\overline{\mathcal{S}}_n} p(s) \le \frac{m(m+1)}{2}. \tag{21}$$

Proof. Fix a source s, and let t be a target that s prefers to its final target. Stability forces

$$\kappa(M^{-1}(t),t) < \kappa(s,t) < \kappa(s,M(s));$$

otherwise (s,t) would block. Distinct preferred targets give distinct matched edges of key strictly smaller than the matched edge of s. Hence

$$p(s) - 1 \le \#\{e \in M : \kappa(e) < \kappa(s,M(s))\}.$$

For $s = s_i$, the right-hand side is at most $i-1$, proving $p(s_i) \le i$. For each i, the first i sources in this ordering provide i proposal counts, all at most i. Hence the ith smallest proposal count is at most i, which is the parking-function condition. Summing the inequalities $p(s_i) \le i$ gives (21). ∎

Example 9.2 (The parking function for $n = 41$). The residual run makes the six proposals

$$s_1 \to t_1, \quad s_2 \to t_1, \quad s_3 \to t_4, \quad s_4 \to t_3, \quad s_2 \to t_4, \quad s_2 \to t_2.$$

Thus

$$p(s_1) = 1, \qquad p(s_2) = 3, \qquad p(s_3) = 1, \qquad p(s_4) = 1.$$

Its nondecreasing rearrangement is $(1, 1, 1, 3)$, and

$$1 \le 1, \qquad 1 \le 2, \qquad 1 \le 3, \qquad 3 \le 4.$$

Thus it is a parking function. The total number of proposals is 6, while Theorem 9.1 gives the bound $4 \cdot 5/2 = 10$. In the control layer this means six generator advances and six proposal-processing steps, of which exactly two involve a key comparison against a held record, illustrating the count used in Theorem 9.5.

Proposition 9.3 (Sharpness). *The bound* (21) *is sharp for the class of globally ranked-pairs instances, even when every key class consists of a single edge.*

Proof. Let the sides be $\{s_1, \dots, s_m\}$ and $\{t_1, \dots, t_m\}$, and give (s_i, t_j) the distinct label $(m + 1)i + j$. Every source orders the targets as $t_1, t_2, \dots, t_m$, and every target orders the sources as $s_1, s_2, \dots, s_m$. In increasing-label greedy order, (s_1, t_1) is selected first; after its endpoints are removed, (s_2, t_2) is selected, and induction gives the diagonal stable matching. The final target t_i has rank i in the list of s_i, so s_i makes exactly i proposals. The total is $m(m + 1)/2$. □

Example 9.4 (Sharpness with three sources). For $m = 3$, all sources rank the targets as t_1, t_2, t_3, and all targets rank the sources as s_1, s_2, s_3. Starting with the queue (s_1, s_2, s_3), source s_1 is held by t_1. Sources s_2 and s_3 are then rejected by t_1; next s_2 is held by t_2, and s_3 is rejected by t_2 before reaching t_3. The proposal counts are therefore

$$p(s_1) = 1, \qquad p(s_2) = 2, \qquad p(s_3) = 3,$$

whose total $6 = 3 \cdot 4/2$ attains the bound.

Theorem 9.5 (Matching-phase operation and auxiliary-storage bound). *Let*

$$N = \sum_{s \in \overline{\mathcal{S}}_n} p(s)$$

be the number of proposals. Conditional on access to the ordered-adjacency generators of Theorem 6.8, the deferred-acceptance control layer makes N generator-advance calls, performs $O(N)$ key comparisons and queue or assignment operations, and stores $O(m)$ auxiliary records outside generator state. In particular,

$$N \le \frac{m(m + 1)}{2}.$$

The stated bound concerns the deferred-acceptance phase only. It does not include the arithmetic work or storage internal to the generators, the enumeration of the residual orbits, or integer bit complexity.

Counted by Theorem 9.5: One generator advance for each proposal; key comparisons; queue and assignment updates; and source, target, and queue records outside the generators.
Not counted by Theorem 9.5: Construction of the residual orbit sets; arithmetic used to produce each generator record; generator-internal storage; integer bit complexity; and the cost of rerunning the computation for an inverse query.

Proof. Every proposal performs at most one key comparison—exactly one when the proposed target is already held, and none when it is free—and a constant number of queue and matching-array updates. Apart from the external generators, the retained state consists of the source status array, one held record per target, and the queue of free sources, all of size $O(m)$. The bound on N is Theorem 9.1. $\square$

Remark 9.6 (On possible subquadratic bounds). Proposition 9.3 shows that an $O(m \log m)$ worst-case bound on the number of proposals cannot follow from the globally-ranked-pairs property alone. Such a bound for the Tunnell-form instances considered here would require additional arithmetic information about the distribution of their midpoint directions.

10. Concluding remarks

The construction brings together a local arithmetic bijection and a global combinatorial completion. Quarter-turns account for congruence-defined subsets, while primitive midpoint directions rank the edges between the remaining orbits. This ranking forces both the unique stable matching and the parking-function bound. The examples for $n = 41$ give one complete fiber of each odd-branch type: one using a quarter-turn and one using the residual stable matching. In both cases the universal even branch supplies the other preimage.

The ordered-adjacency generator avoids storing the complete $m \times m$ edge table. It would be interesting to determine whether the arithmetic distribution of midpoint directions yields a better complexity bound for these Tunnell-form instances, or whether the residual partner admits a direct description from one representation.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the mathematical development and preparation of this work, the authors used OpenAI's ChatGPT and Codex through an AI-assisted solver–referee pipeline for symbolic experimentation, literature searches, proof organization, and language revision. The authors reviewed and edited the resulting content, checked the mathematical arguments and cited sources, and take full responsibility for the article.

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Department of Mathematics, The University of Alabama, Tuscaloosa, AL 35487, USA
Email address: `agurung3@crimson.ua.edu`

Department of Mathematics, The University of Alabama, Tuscaloosa, AL 35487, USA
Email address: `klee94@ua.edu`